CHARAN TEJA VALAVALA

+91 92476 15799 | f20220059@hyderabad.bits-pilani.ac.in | Hyderabad, India
GitHub | Portfolio | LinkedIn

SOFTWARE ENGINEERING | APPLIED AI

SELECTED PROJECTS

InspectLab | Visual Defect Detection | Code | Demo — Python | DINOv2 | ONNX

- Built visual defect detection with **pretrained DINOv2 vision-transformer patch features**, cosine nearest-reference scoring, heatmaps and separate normal-only threshold calibration.
- Compared 1/4/16 references over **349 MVTec test images**, 3 categories and 3 seeds: mean image AUROC **0.988 vs 0.841** RGB baseline at 16 references; pixel AUROC 0.945 at 56 x 56.

Course Copilot | Hybrid Retrieval | Code | Demo — Python | FastAPI | MiniLM | SQLite

- Built PDF/TXT ingestion, file/page citations, MiniLM + BM25 rank fusion and an automated NLI support gate for optional generated answers.
- Evaluated **300 public SciFact queries over 5,183 abstracts**: Hit@5 **78.7%** vs 76.0% lexical; MRR@5 0.668.

IncidentPilot | Incident Investigation | Code | Demo — Python | scikit-learn | Qwen

- Built read-only tools, evidence checks and approval-gated simulated actions. A classifier trained on 20 synthetic cases scored **8/8** on separate stress cases vs **4/8** for local Qwen scoring; limited to four known causes.

FormScope | Financial Data & Search | Code | Demo — Python | FastAPI | ONNX | XBRL

- Implemented exact-period XBRL joins, restatement handling and lexical + MiniLM filing search. Matched **40/40** revenue/net-income values against pinned references from 20 filings; linked ratios to source facts.

ACADEMIC PROJECTS

Diabetic Retinopathy Detection — Python | OpenCV | SVM

- Extracted retinal-image color/texture features, trained SVM classifiers and evaluated accuracy, precision, recall and sensitivity.

Reinforcement Learning for Object Tracking — Python | Monte Carlo control

- Built a stochastic grid with terminal states and teleportation; compared on-policy and off-policy Monte Carlo control.

EXPERIENCE & EARLY-STAGE WORK

MDDTI SBC | Web Development Intern — May - July 2024

- Contributed HTML/CSS pages and MongoDB-backed features to a website and online test platform used by **200+ students and trainees**.

DataForge | Founder, concept-stage venture — February - May 2026

- Explored an AI infrastructure and data center venture with mentor guidance and early funding discussions; remained at the concept stage.

EDUCATION & ACHIEVEMENT

BITS Pilani, Hyderabad Campus | B.E. Computer Science Engineering (currently enrolled)

AP EAMCET: Statewide rank 8

Narayana Junior College | Class XII: 98.1% | 2022

Narayana High School | Class X: 10/10 GPA | 2020

TECHNICAL SKILLS & COURSEWORK

Programming: Python, SQL, C, C++, Java, JavaScript, HTML, CSS

Data and ML: NumPy, Pandas, Matplotlib, scikit-learn, OpenCV, PyTorch, Hugging Face Transformers

Applied AI: DINOv2, embeddings, BM25, retrieval evaluation, ONNX, Sentence Transformers, Ollama

Backend and delivery: FastAPI, REST APIs, MongoDB, SQLite, Git, Docker, pytest, GitHub Actions

Coursework: Artificial Intelligence, Machine Learning, Reinforcement Learning, Foundations of Data Science, Data Structures and Algorithms, Database Systems, Object-Oriented Programming

Languages: English, Hindi, Telugu